

Muhammad Zain Ali (AI/NLP Researcher)

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📍 Hamilton, New Zealand | Open to Relocation

Profile Summary

AI/NLP researcher completing a Ph.D. in Computer Science at the University of Waikato, specializing in large language models (LLMs), low-resource and multilingual NLP, and misinformation detection. Published at leading international venues, including ACL and ICONIP, with experience spanning academic research, teaching, and industry NLP development.

Experience

- 2023 – Present 📌 **Teaching Assistant, University of Waikato**
📍 *Hamilton, New Zealand*
- Provide teaching and laboratory support for undergraduate courses in Machine Learning, Web Development, and Engineering Computing.
 - Support students with machine learning practicals, programming, and computational concepts through lab demonstrations and assessment.
- 2022 – 2023 📌 **Software Engineer – NLP, Cityscape Technology Pvt. Ltd.**
📍 *Lahore, Pakistan*
- Developed NLP components for a virtual assistant targeting the Canadian real-estate market, including question answering and conversational interfaces.
 - Developed place-intelligence and mortgage-affordability components to support property search and real-estate decision-making.
- 2020 – 2022 📌 **Research Engineer – ML/DL/NLP, Center for Language Engineering**
📍 *Lahore, Pakistan*
- Conducted research and development in Urdu NLP, including sentiment analysis and political aspect-based sentiment analysis.
 - Contributed to NLP applications including the Parrakh plagiarism detection system and Roman-Urdu-to-Urdu language processing.
- 2016 – 2020 📌 **Assistant Manager, Pakistan Telecommunication Company Limited (PTCL)**
📍 *Lahore, Pakistan*
- Worked across network planning, network-health analysis, GIS-based inventory management, GPON deployment, and network transformation.
 - Supported operational decision-making through data analysis and workforce and network performance monitoring.

Education

- 2023 – Present 📌 **Ph.D. Computer Science, University of Waikato**
📍 *Hamilton, New Zealand*
Thesis: *Enhancing Misinformation Detection with a Focus on Urdu Language.*
Status: Thesis submitted and under examination.
- 2017 – 2020 📌 **M.Sc. Electrical Engineering, University of Engineering & Technology**
📍 *Lahore, Pakistan*
Thesis: *Use of Social Media for Epidemic Surveillance.*
- 2012 – 2016 📌 **B.Sc. Electrical Engineering, University of Engineering & Technology**
📍 *Lahore, Pakistan*
Final Year Project: *Electronic Biometric Voting System.*

Selected Publications

- 1 **M. Z. Ali**, Y. Wang, M. A. Manzoor, T. Smith, and B. Pfahringer, “Markhor: A framework for adapting LLMs to Urdu via alignment-free tokenizer replacement,” in *Findings of the Association for Computational Linguistics: ACL-IJCNLP*, 2026.
- 2 **M. Z. Ali**, T. Smith, and B. Pfahringer, “Fake news classification in Urdu through domain adaptation of multilingual language models,” in *33rd International Conference on Neural Information Processing*, Springer, 2026.  URL: <https://arxiv.org/abs/2512.22778>
- 3 **M. Z. Ali**, Y. Wang, B. Pfahringer, and T. C. Smith, “Detection of human and machine-authored fake news in Urdu,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Vienna, Austria: Association for Computational Linguistics, Jul. 2025, pp. 3419–3428.  DOI: 10.18653/v1/2025.acl-long.170
- 4 **M. Z. Ali**, T. Smith, and B. Pfahringer, “Enhanced stance detection using cascaded siamese networks with attention mechanism,” in *31st International Conference on Neural Information Processing*, Springer, 2024, pp. 377–392.  URL: https://link.springer.com/chapter/10.1007/978-981-96-6599-0_26
- 5 **M. Z. Ali**, E. Haq, S. Rauf, K. Javed, and S. Hussain, “Improving hate speech detection of Urdu tweets using sentiment analysis,” *IEEE Access*, pp. 84 296–84 305, 2021.  DOI: 10.1109/ACCESS.2021.3087827
- 6 **M. Z. Ali**, K. Javed, E. Haq, and A. Tariq, “Sentiment and emotion classification of epidemic related bilingual data from social media,” *arXiv preprint*, 2021.  URL: <https://arxiv.org/abs/2105.01468>

Technical Expertise

Machine Learning & NLP	 Large Language Models (LLMs), Transformers, Multilingual and Low-Resource NLP, Domain Adaptation, Fine-Tuning, Model Evaluation, Text Classification, Question Answering.
Programming & ML Frameworks	 Python, PyTorch, TensorFlow, Hugging Face, Scikit-learn, Pandas, NumPy, Matplotlib.
Development & Research Tools	 Flask, REST APIs, Git, $\text{\LaTeX}$, MATLAB, MySQL, HTML/CSS.
Cloud Platforms	 Google Cloud Platform (GCP), Microsoft Azure.

Selected Professional Training

- 2025  **Generative AI with Diffusion Models**, NVIDIA.
- 2022  **Machine Learning on Google Cloud (Vertex AI and AI Platform)**, Udemy.
-  **Natural Language Processing with Python**, Udemy.
-  **Natural Language Processing Specialization**, Coursera.
- 2021  **Reinforcement Learning Foundations**, LinkedIn Learning.
-  **SQL + Tableau**, 365 Data Science.

Academic Referees

Prof. Bernhard Pfahringer
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Dr. Yuxia Wang
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